

COM738 – MSc Dissertation (Level 7)

Welcome to this essential session on the MSc Dissertation, a cornerstone of your Level 7 computing studies. This lecture aims to demystify the dissertation process, equipping you with the knowledge and strategies to excel in this significant academic undertaking.



Module Overview: COM738 Dissertation

Module Code

COM738

Module Title

Dissertation

Level

7 (MSc)

Credit Value

60 Credits

Total Study Hours

600 hours

The COM738 Dissertation module stands as the most substantial academic component of your MSc programme. It challenges you to independently conceive, execute, and present a significant piece of research or development. This endeavour is designed to demonstrate your Master's-level knowledge, foster originality in your work, and cultivate a professional approach to complex problem-solving.

An impressive **594 hours** are dedicated to guided independent study, reflecting the profound depth and autonomy expected in this module.

Lecture Purpose: Guiding Your Dissertation Journey

This focused session is meticulously designed to provide clarity and direction for your dissertation. We will cover critical aspects to ensure you are well-prepared for the challenges and opportunities ahead:

01

Setting Expectations

Preparing students for the rigour and **expectations of a Level 7 dissertation.**

02

Dissertation Types

Clarifying the differences between **research-based and standard (project-based) dissertations.**

03

Assessment & Milestones

Explaining the **assessment structure, key milestones, and deadlines.**

04

Key Processes

Guiding students on **proposal development, supervision protocols, and effective academic writing.**

05

Examiner Insights

Helping students understand **what examiners look for in a high-quality MSc dissertation.**

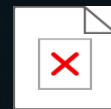
Learning Outcomes Alignment: Your Dissertation's Impact

By the conclusion of this lecture, you will possess a clear understanding of how to effectively address the key learning outcomes of the COM738 module. These outcomes are fundamental to producing a robust and successful dissertation:



Research Methodologies (LO1)

Apply appropriate **research methodologies** to complex computing problems.



Originality & Professionalism (LO2)

Demonstrate **originality** in thought and a professional approach to project execution.



Critical Evaluation (LO3)

Critically evaluate existing literature and analyse gathered data with rigor.



Knowledge Advancement (LO4)

Contribute to the advancement of subject knowledge within the computing discipline.



Academic Structure (LO5)

Structure and write a substantial academic document of Master's quality.

Understanding the MSc Dissertation

What is an MSc Dissertation?

An MSc dissertation is an **independent, self-directed investigation** into a computing-related problem or research question. It is not merely a taught module; rather, it reflects your cultivated ability to:

Identify Problems

Pinpoint a meaningful and relevant problem within your field.

Systematic Investigation

Investigate the problem using a structured and systematic approach.

Apply Theory

Apply theoretical knowledge to practical scenarios and real-world challenges.

Critical Evaluation

Critically evaluate the outcomes and implications of your work.

At Level 7, your work must transcend mere **implementation**. Simply building a system is insufficient; you must rigorously justify your design decisions, evaluate alternative approaches, and critically reflect on the entire process. This holistic view is what distinguishes Master's-level work.

Dissertation Types & Research Expectations

Two Categories of Dissertation

1. Research-Based Dissertation

Focuses on **investigating a specific research problem**, generating new knowledge or understanding.

- Clear research questions or hypotheses.
- Systematic literature review to contextualise the work.
- Use of qualitative, quantitative, or mixed research methods.
- Rigorous data collection and analysis.
- Critical discussion of findings and their implications.

Examples: AI effectiveness in cyber forensics, security vulnerabilities in cloud computing, ethical implications of machine learning systems.

2. Standard (Project-Based) Dissertation

Focuses on **substantial technical development** coupled with robust academic evaluation.

- Detailed software or system design and implementation.
- Clear justification of chosen tools and technologies.
- Comprehensive testing and evaluation strategies.
- Critical reflection on system performance and inherent limitations.

Examples: Secure mobile application development, network intrusion detection system, data analytics platform.

❏ **⚠️ Important:** Even a standard dissertation **must still include research, thorough evaluation, and critical analysis** to meet Level 7 academic standards. It is not merely a development project.

Proposal, Supervision & Ethics

Research Proposal (10% Assessment)

Your research proposal, with an **indicative length of 2,000 words**, is a crucial first step. Its purpose is to demonstrate the feasibility, academic value, and ethical awareness of your intended study.

1

Title & Context

Clear title, background, and problem statement.

2

Aim & Objectives

Precisely defined research aim and measurable objectives.

3

Literature & Methodology

Preliminary literature review and proposed methodology.

4

Ethics & Outcomes

Ethical considerations, expected outcomes, and a project timeline.

Dissertation Approval Panel

Before commencing your full dissertation work, your proposal will undergo review by a Dissertation Approval Panel. This panel will assess:

- The academic rigour and feasibility of your proposal.
- Ethical and professional issues pertinent to your research.
- The suitability of your proposed supervisor, ensuring alignment with your project.

Ethics and Professional Responsibility

All students must diligently consider the ethical implications of their research, including data privacy and consent, use of human participants, security of collected data, and legal compliance. Failure to adequately address these ethical considerations may lead to **proposal rejection or even dissertation failure**.

Writing, Assessment & Examiner Expectations

Dissertation Assessment (90%)

The primary assessment for your dissertation accounts for 90% of the module grade. You have two distinct options:

Typical Dissertation Structure

- Abstract, Introduction, Literature Review
- Methodology, Implementation / Investigation
- Results and Analysis, Discussion
- Conclusion and Future Work, References, Appendices

What Examiners Look For

Examiners seek specific hallmarks of quality in a Master's dissertation:

- ✓ Critical Thinking: Demonstrates analysis, not just description.
- ✓ Justified Decisions: All choices are logically supported.
- ✓ Theory-Practice Link: Strong connection between academic theory and practical application.
- ✓ Originality: Evidence of unique contributions or innovative approaches.
- ✓ Academic Writing: High quality, clear, and concise academic prose.
- ✓ Referencing: Correct and consistent citation throughout.

Common Reasons for Low Marks

- Weak or outdated literature review.
- Poor methodology justification.
- Lack of rigorous evaluation.
- Overly descriptive, lacking critical insight.
- Ineffective time management.
- Ignoring supervisor feedback.



Option 1: Written Dissertation

A comprehensive document of 15,000–20,000 words.



Option 2: Academic Paper

A maximum 17-page paper, suitable for publishable academic standards.



Guided Independent Study Expectations

Your MSc Dissertation journey is predominantly an exercise in independent learning. To succeed, you are expected to maintain consistent engagement and proactive management of your research:

1

Consistent Work

Work consistently throughout the semester, dedicating regular hours to your project.

2

Regular Supervision

Attend and prepare for regular meetings with your supervisor, typically on a weekly basis.

3

Progress Tracking

Maintain a detailed research log or progress record to track your development and findings.

4

Continuous Development

Actively demonstrate continuous development of your research skills and project.

❏ **Remember:** 600 hours of study approximates a full-time academic workload for a significant portion of the academic year. Plan your time effectively.

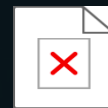
Employability & Graduate Attributes

Successfully completing your dissertation develops a suite of highly sought-after skills, directly enhancing your employability and equipping you with essential graduate attributes:



Critical Thinking

Enhanced analytical and problem-solving abilities.



Research Skills

Proficiency in identifying, investigating, and synthesising information.



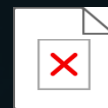
Professional Communication

Ability to articulate complex ideas clearly and persuasively.



Ethical Awareness

Understanding of responsible conduct in research and practice.



Independent Working

Cultivation of self-reliance, initiative, and resilience.

These skills are invaluable for a range of future pathways:

PhD Study

Research Roles

Senior Technical
Positions

Industry Innovation